

CSE 4616 – Wireless Networks

Lab 01: In-Lab Assessment Tasks

Duration: 1 hour 30 minutes — Individual Assessment

Tools: OMNeT++ and INET Framework

General Instructions

- All tasks are based on the **prebuilt sample** network `WiredAndWirelessHostsWithAP` studied in the lab manual.
- Each task builds on the original NED and INI files. Unless stated otherwise, begin from the original files and apply the required modifications.
- Read each task fully before starting. Pay attention to the **Deliverables** — submit exactly what is listed.
- Name your files clearly:
`TaskX_YourStudentID.ned` and `TaskX_YourStudentID.ini`
- Take Qtenv screenshots *while the simulation is running* (packets visible) unless the task states otherwise.
- Write your explanations in the provided answer sheet or as comments inside the submitted files.

Important Note

The **original** network has:

- 3 hosts: `wiredHost1`, `wiredHost2`, `wirelessHost1`
- 1 router, 1 access point
- 3 apps per host: `app[0] = UdpEchoApp` (port 1000), `app[1]` and `app[2] = UdpBasicApp`
- Packet size: 100 B, send interval: 1 s, stop time: 300 s, simulation limit: 400 s

Task 01: Modify Traffic Parameters

Background: In the original simulation, each host sends 100 B packets every 1 second and stops at 300 s. In this task, you will modify the traffic parameters in the INI file to simulate a **higher-load** scenario and observe the difference in Qtenv.

Required Modifications (INI file only):

1. Change the **packet size** from 100B to 500B for all sending apps.
2. Change the **send interval** from 1s to 0.5s for all sending apps (i.e., send twice as fast).
3. Change the **stop time** of all sending apps from 300s to 350s.
4. Change the overall **simulation time limit** from 400s to 500s.

Analysis Questions (write your answers in the report):

- a. How many packets per second does each host generate in total after your modification? Show your calculation.
- b. What is the total data rate (in bytes/second) generated by `wiredHost1` after the modification?
- c. In Qtenv, did you notice any visual difference compared to the original simulation? Describe what you observed.

Deliverables

- **Modified INI file:** `Task1_StudentID.ini`
- **Screenshot:** Qtenv window showing the simulation running with packets visible (taken between $t = 50$ s and $t = 200$ s)
- **Written answers** to analysis questions (a), (b), and (c)

Task 02: Add a Second Wireless Host

Background: The original network has only one wireless host (`wirelessHost1`). In this task, you will add a **second wireless host** (`wirelessHost2`) to the NED file and update the INI file so that all hosts exchange traffic with each other.

Required Modifications:

NED file:

1. Add a new submodule `wirelessHost2` of type `WirelessHost` to the network. Place it at display coordinates `@display("p=62,140")`.
2. No new cable connections are needed — `wirelessHost2` connects to the access point **wirelessly** through the existing `radioMedium`, just like `wirelessHost1`.

INI file:

1. The line `**.*Host*.numApps = 3` already applies to `wirelessHost2` automatically. Verify this and state why in your answer sheet.
2. Add destination address lines so that:
 - `wirelessHost2`'s `app[1]` sends to `wiredHost1`
 - `wirelessHost2`'s `app[2]` sends to `wirelessHost1`
3. Update `wiredHost1`'s `app[2]` destination from `"wiredHost2"` to `"wirelessHost2"` so that it now sends to the new wireless host instead.

Analysis Questions:

- a. Draw the updated traffic map (who sends to whom) after your modifications. Use arrows between host names.
- b. Why does `wirelessHost2` not need a new cable connection in the NED file? Explain the role of `radioMedium` in your answer.
- c. In Qtenv, can you identify `wirelessHost2` on the canvas? Describe its position relative to the access point.

Deliverables

- **Modified NED file:** `Task2_StudentID.ned`
- **Modified INI file:** `Task2_StudentID.ini`
- **Screenshot:** Qtenv canvas showing all 4 hosts (both wireless hosts visible near the access point)
- **Written answers** to analysis questions (a), (b), and (c)

Task 03: Add a Third Wired Host and a New Application

Background: In this task you will extend the wired side of the network by adding a **third wired host** (wiredHost3) connected directly to the router. You will also increase the number of apps per host to 4 and configure the new app[3] on each host to send traffic to wiredHost3.

Required Modifications:

NED file:

1. Add a new submodule wiredHost3 of type StandardHost at display position `@display("p=412,140")`.
2. Connect wiredHost3 to the router using an Eth100M cable:

```
1 wiredHost3.ethg++ <--> Eth100M <--> router.ethg++;
```

INI file:

1. Change numApps from 3 to 4 for all hosts.
2. app[0] remains UdpEchoApp on port 1000 (no change needed — explain why in your answer).
3. app[1] and app[2] keep their original destinations (no change needed).
4. Configure app[3] on **every host** to send to wiredHost3 with:
 - destPort = 1000
 - messageLength = 200B
 - sendInterval = 2s
 - stopTime = 300s
5. Set wiredHost3's app[1] destination to "wiredHost1" and app[2] to "wirelessHost1". Set app[3] to "wiredHost2".

Analysis Questions:

- a. The line `**.*Host*.app[0].typename = "UdpEchoApp"` was not changed, yet app[0] on wiredHost3 is automatically set to UdpEchoApp. Why does this happen? Reference the wildcard and specificity rules in your answer.
- b. What is the total number of packets per second now generated across the entire network? Show your working.
- c. In Qtenv, describe the path a packet takes from wiredHost3 to wirelessHost1. Name every intermediate device it passes through.

Deliverables

- **Modified NED file:** `Task3_StudentID.ned`
- **Modified INI file:** `Task3_StudentID.ini`
- **Screenshot:** Qtenv canvas clearly showing `wiredHost3` connected to the router
- **Written answers** to analysis questions (a), (b), and (c)

Task 04: Staggered Traffic and Selective Recording

Background: In a realistic network, not all devices start transmitting at the same time. In this task you will introduce **staggered start times** for each host's sending apps, change the **send interval distribution** from deterministic to random, and **selectively enable vector recording** for only one host and the router to limit result file size.

Required Modifications (INI file only):

1. **Disable global vector recording** and enable it only for wiredHost1 and the router:

```
1 **.vector-recording = false
2 **.wiredHost1*.vector-recording = true
3 **.router*.vector-recording = true
```

2. **Stagger the start times** of the sending apps so that hosts begin transmitting at different simulated times:

```
1 **.wiredHost1.app[1].startTime = 0s
2 **.wiredHost1.app[2].startTime = 5s
3 **.wiredHost2.app[1].startTime = 10s
4 **.wiredHost2.app[2].startTime = 15s
5 **.wirelessHost1.app[1].startTime = 20s
6 **.wirelessHost1.app[2].startTime = 25s
```

3. **Change the send interval** of all sending apps from a fixed 1s to a random exponential distribution with mean 1s:

```
1 **.*Host*.app[1..].sendInterval = exponential(1s)
```

4. Keep all other parameters (packet size, stop time, simulation limit) at their **original values**.

Analysis Questions:

- a. At simulated time $t = 12\text{s}$, which hosts are actively sending packets and which are not? Justify your answer using the start times you configured.
- b. Why is `exponential(1s)` a more realistic model for network traffic than a fixed 1s interval? What real-world phenomenon does this distribution model?
- c. Why is it useful to disable vector recording globally and enable it only for selected modules? What problem does this solve in large simulations?
- d. In Qtenv, run the simulation and take a screenshot at exactly $t = 22\text{s}$ (use the step/pause controls). Which hosts should be visible as active senders at that moment? Does your screenshot confirm this?

Deliverables

- **Modified INI file:** `Task4_StudentID.ini` (NED file is unchanged from the original)
- **Screenshot 1:** Qtenv canvas paused at approximately $t = 12$ s showing which hosts are active
- **Screenshot 2:** Qtenv canvas paused at approximately $t = 22$ s showing all hosts active
- **Written answers** to analysis questions (a), (b), (c), and (d)